

DEPI – Software Testing Track

Part 01: Java Fundamentals

Instructor: Mina Younan

Session 08 – Task 3

Question 1: Java Collections – Managing a List of Items

Task

Create a Java Maven project that manages a collection of items using the different data structures and collection types studied in Lecture 08-1.

Your project should demonstrate how the same item data can be stored, accessed, searched, modified, and displayed using different Java collection types.

You should use appropriate examples of:

- One-dimensional array (Array)
- Two-dimensional array (2D Array)
- List
- ArrayList
- LinkedList
- Set
- HashSet
- HashMap

You do not need to use all collection types for the same purpose. Choose an appropriate data structure for each task and explain why you selected it.

Example: Two-Dimensional Array

Create a 2D array to store basic item information such as:

ID	Price
101	500
102	700
103	25000
104	7000

Example:

```
double[][] itemData = { {101, 500}, {102, 700}, {103, 25000}, {104, 7000} };
```